

INPUT & OUTPUT DEVICES PRACTICE SHEET

NIMCET TARGET PAPER • COMPUTER BASICS (MEDIUM LEVEL)

ResourceOS

1. In Memory-Mapped I/O addressing, which of the following statements is TRUE?

- (A) Special I/O instructions like IN and OUT are strictly required.
- (B) The I/O devices share the same address space with memory.
- (C) Control lines for I/O and Memory are physically separate.
- (D) Maximum memory addressable capacity is increased.

2. What primary advantage does Isolated I/O offer over Memory-Mapped I/O?

- (A) Simpler arithmetic and logical operations on I/O data
- (B) Preserves the full memory address space for main memory RAM
- (C) Requires fewer control lines on the system bus
- (D) Faster execution of I/O transfer loops

3. In Direct Memory Access (DMA), what occurs during Cycle Stealing mode?

- (A) The DMA controller takes control of the bus for an entire block transfer.
- (B) The DMA controller transfers one word and yields bus control back to CPU.
- (C) Transfers occur only when the CPU is executing internal non-bus instructions.
- (D) The CPU is completely suspended until I/O finishes completely.

4. How does Vectored Interrupt servicing differ from Non-Vectored Interrupt servicing?

- (A) In Vectored Interrupts, the CPU polls all devices to identify the interrupt source.
- (B) In Vectored Interrupts, the interrupting device supplies the interrupt vector address.
- (C) Non-Vectored interrupts have higher priority than vectored interrupts.
- (D) Non-Vectored interrupts use a dedicated hardware interrupt controller.

5. In priority interrupt handling, what is a characteristic feature of the Daisy Chaining method?

- (A) All devices are connected in parallel to an interrupt encoder.
- (B) Interrupt acknowledge signal propagates serially through devices in priority order.
- (C) Software polling determines device priority dynamically.
- (D) Highest priority is given to the device furthest from the CPU.

6. What is the fundamental operational distinction between Vector (Random Scan) and Raster Scan displays?

- (A) Vector displays scan line-by-line from top to bottom across the whole screen.
- (B) Vector displays draw lines directly from point to point as directed by line commands.
- (C) Raster displays do not use a frame buffer memory array.
- (D) Raster displays can only display monochrome vector graphics.

7. Calculate the frame buffer memory size required for a 1024 × 768 display with 24-bit True Color depth.

- (A) ~786 KB
- (B) ~2.25 MB
- (C) ~18.87 MB
- (D) ~2.36 MB

8. Which signal is issued by the DMA controller to request control of the system bus from the CPU?

- (A) HOLD (or BUS REQUEST)
- (B) HLDA (BUS ACKNOWLEDGE)
- (C) INTERRUPT REQUEST (IRQ)
- (D) READ/WRITE CONTROL

9. In asynchronous serial data transmission, what purpose do 'Start' and 'Stop' bits serve?

- (A) Error detection and parity calculation
- (B) Frame synchronization between sender and receiver clocks
- (C) Data compression and flow control
- (D) Device address decoding

10. Which handshaking mechanism ensures asynchronous data transfer reliability without a shared clock?

- (A) Strobe control signal alone
- (B) Two-wire or multi-wire handshake (Data Ready / Data Accepted)
- (C) Synchronous bus clock pulsing
- (D) DMA burst mode streaming

11. An I/O port has 8 data lines. If a transfer takes place at 115,200 Baud with 1 start bit, 8 data bits, and 1 stop bit, what is the effective data transfer rate?

- (A) 14.4 KB/s
- (B) 11.52 KB/s
- (C) 115.2 KB/s
- (D) 9.6 KB/s

12. What is the main operational difference between an I/O Interface and an I/O Processor (Channel)?

- (A) An I/O Interface can execute its own instruction programs stored in memory.
- (B) An I/O Processor has its own instruction set and can execute complete I/O programs independently.
- (C) An I/O Interface connects directly to the CPU registers without a system bus.
- (D) An I/O Processor operates strictly in Programmed I/O mode.

13. What occurs during the 'Interlaced Scanning' technique used in display monitors?

- (A) All scan lines are drawn sequentially in a single vertical pass.
- (B) Odd-numbered scan lines are drawn first, followed by even-numbered lines in two passes.
- (C) Pixels are drawn vertically from left to right across columns.
- (D) Screen refresh is performed only when the image content changes.

14. In a Dot Matrix Printer, how is character formation achieved?

- (A) By pressing pre-formed character molds against an ink ribbon
- (B) By firing vertical arrays of print pins striking an ink ribbon against paper
- (C) By using electrostatic charges on a photosensitive drum
- (D) By spraying thermal dyes using heating elements

15. What is the primary cause of CPU efficiency loss in Programmed I/O with Polling?

- (A) Frequent hardware interrupt context switching overhead
- (B) CPU spinning in a tight loop repeatedly checking status bits of slow devices
- (C) Bus contention caused by DMA controllers
- (D) High memory footprint of interrupt vector tables

16. Which register in an I/O interface chip holds status information such as BUSY, READY, and ERROR bits?

- (A) Data Buffer Register
- (B) Control Register
- (C) Status Register
- (D) Address Line Register

17. In an interrupt-driven I/O system, what is the sequence of actions executed by the CPU upon receiving an interrupt?

- (A) Execute ISR → Save Context → Fetch Vector → Restore Context
- (B) Complete current instruction → Save Program Counter/Context → Fetch ISR address → Execute ISR → Restore Context
- (C) Halt CPU immediately → Reset registers → Execute ISR → Jump to origin
- (D) Transfer bus control to DMA → Execute ISR → Return to CPU instruction

18. What is the function of a Command Register within an I/O Interface module?

- (A) To store incoming raw data received from the external peripheral
- (B) To store control commands/mode specifications sent by CPU to configure the device
- (C) To generate clock synchronization signals for data lines
- (D) To hold memory addresses for direct bus operations

19. How does a Capacitive Touchscreen detect user touch inputs?

- (A) By measuring pressure physical contact flexing flexible glass layers
- (B) By detecting changes in electrical capacitance caused by human finger conductivity
- (C) By blocking grid infrared light beams above screen surface
- (D) By surface acoustic wave reflection timing

20. In DMA transfer modes, what characterizes "Transparent Mode" (Hidden DMA)?

- (A) DMA transfers data only when CPU is executing instructions that do not use the system bus
- (B) CPU transfers control of all system buses for several seconds
- (C) Data is transferred directly between two I/O devices without memory
- (D) Transfers are forced via high-priority hardware interrupts

21. What is the role of a Frame Buffer (Video RAM) in modern graphics systems?

- (A) Store CPU microcode for rendering 3D transformations
- (B) Store pixel intensity and color bit values for screen rendering
- (C) Buffer incoming keyboard/mouse input commands
- (D) Store printer spool files before output

22. Why are I/O Interface modules required between peripherals and the CPU system bus?

- (A) Peripherals use identical signal levels and speeds as the CPU bus.
- (B) Peripherals operate at different speeds, data formats, and signal standards than CPU.
- (C) To prevent RAM memory from being accessed by non-CPU components.
- (D) Peripherals cannot transmit digital data without interface translation.

23. What is the advantage of an Optical Mouse over a traditional Mechanical Ball Mouse?

- (A) Higher mechanical friction providing better physical grip
- (B) No moving mechanical parts, reducing wear and tracking errors from dust
- (C) Lower digital sampling frequency requirement
- (D) Direct compatibility with non-reflecting transparent glass without sensor

24. In bus arbitration, what distinguishes Centralized Parallel Arbitration from Daisy Chaining?

- (A) Centralized Parallel Arbitration uses independent request and grant lines for each device.
- (B) Daisy Chaining offers equal priority to all requesting devices.
- (C) Centralized Parallel Arbitration requires fewer bus request lines.
- (D) Daisy Chaining is faster due to direct hardware encoding.

25. Which optical scanning technology uses pattern recognition algorithms to read full formatted text documents?

- (A) OMR (Optical Mark Recognition)
- (B) OCR (Optical Character Recognition)
- (C) MICR (Magnetic Ink Character Recognition)
- (D) Barcode Reader

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(B)	11	(B)	16	(C)	21	(B)
2	(B)	7	(D)	12	(B)	17	(B)	22	(B)
3	(B)	8	(A)	13	(B)	18	(B)	23	(B)
4	(B)	9	(B)	14	(B)	19	(B)	24	(A)
5	(B)	10	(B)	15	(B)	20	(A)	25	(B)